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Constant current charging time based fast state-of-health estimation for lithium-ion batteries

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ABSTRACT

The state of health (SOH) estimation is critical for a battery management system's safe operation. Considering feature extraction, time-consuming, model/calculation complexity problems, a battery SOH estimation method based on constant current charging time (CCCT) is proposed in this paper. Unlike previous works, it is proved that CCCT can perfectly replace incremental capacity peak area. Since no filtering process is required in this method, the validity of the feature is maximally preserved. The random forest regression is combined to form accurate and fast SOH estimation. The proposed method is validated with the Oxford and CALCE datasets, collected from different batteries under different conditions. The average root-mean-square error of 8 cells for SOH estimation is 0.52%. Compared with the incremental capacity analysis (ICA)-based SOH estimation method, the prediction accuracy of the proposed method is improved by 41.6%, and fewer data are utilized. Besides, the time needed for the model training and prediction of the proposed method is less than 1 s. Additionally, the proposed method is proved to have good adaptability to different voltage ranges and charging/discharging conditions.

1. Introduction

Lithium-Ion batteries are widely used in electric vehicles (EV), mobile robots, and grid energy storage systems (ESS) [1–3]. The battery state of health (SOH) is usually defined as the percentage between the current available capacity and the initial capacity of the battery. Due to the complex aging mechanism of the lithium-ion battery, an effective method for estimating SOH is particularly important, directly affecting the battery's performance, safety, and reliability in the working process [4]. Many methods have been proposed to estimate the SOH, which can be divided into three categories: 1) The direct measurement [5,6]; 2) The model based method [7,8]; 3) The data driven method [9,10]. For the first two methods, stable experimental environments or complex test conditions are needed, limiting their application. The data-driven method has become a research hotspot of SOH estimation for its flexible and universal characteristics in recent years. It is mainly concerned with two things: one is to extract the features related to battery aging from the charging or discharging processes. These features are usually called health indicators (HIs). Another is to use machine learning (ML) models to fit the linear or nonlinear relationship between HIs and SOH, and predict SOH on new test cases. The Gaussian process regression (GPR) [11,12], support vector machine (SVM) [13,14], random forest (RF) [15–17], neural network (NN) [18,19], and other ML models have been widely used in the field of SOH estimation, and have achieved good results in estimation accuracy.

Multiple health indicators combined with data-driven technology are used to estimate battery SOH. Meng et al. [14] used the voltage response of short-term pulse current as the HIs and used the integrated SVM to estimate SOH. Kong et al. [1] combined the features of differential voltage and surface temperature and obtained a good SOH estimation effect on the fast charging dataset using GPR. From the perspective of electrochemical impedance spectroscopy (EIS), Fu et al. [9] proposed a fast SOH estimation method based on impedance features and ELM, which overcomes the influence of fluctuation on traditional measurement signals. Hoque et al. [20] found that the internal resistance dynamics of the battery is closely related to the battery capacity to establish a health prediction model and identify the consistency of the battery at an early stage. Son et al. [21] extracted multi-physical features as

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health indicators from the perspective of mechanical pressure and equivalent impedance to explain the multi-physical degradation mechanism of the battery. In addition to studying health features from multiple perspectives, Lin et al. [22] also fused multiple machine learning models to estimate battery SOH, which achieved higher robustness and estimation accuracy. The above studies have established feature-based SOH prediction models from different angles. However, due to the influence of actual operating conditions and measurement noise, the availability of these health indicators is challenged, and it is often difficult to obtain an ideal effect in practice.

Battery charge or discharge curves are usually the easiest to obtain, and many studies directly extracted HIs from these curves to estimate SOH. Richardson et al. [23] used the time values at equispaced voltages as the inputs to GPR to estimate SOH, but a Savitzky−Golay (SG) filter is needed for smoothing the voltage curve. Wang et al. [7] discovered that the charging current curve during the constant voltage profile was related to the battery SOH, and the constant voltage charging aging factor is recognized as a new effective HI. However, this feature extraction and transformation process is very time-consuming. Liu et al. [24] used the voltage variation during equal time intervals to estimate SOH. Although good results are obtained, there is almost no fixed discharge pattern in the actual operation of the battery.

In addition, incremental capacity analysis (ICA) is considered as a mature technology in SOH estimation. Various features such as peak area [25,26], peak height, and peak position [27,28] are extracted from IC curves to estimate SOH. Compared with the voltage curves, the IC curves are usually more interpretable for battery aging because these features on IC curves are very sensitive to the specific electrochemical reaction of the battery [29]. These IC peaks have been proved to be closely related to the phase transition phenomena of active materials during the Li-ion intercalation/deintercalation process [30]. Li et al. [12] used an advanced Gaussian filter method to smooth the IC curves, and based on the 11 sampling points of the IC curves combined with GPR to estimate the SOH. Tian et al. [31] took the part of the differential temperature curves and combined it with the ICA method to improve the SOH prediction accuracy. In particular, in recent research, Ospina Agudelo et al. [25] proved that the main peak area is the optimal feature on the IC curves to evaluate battery SOH under high current tests and random usage patterns. Although it is effective to extract features from IC curves for SOH estimation, there are still many shortcomings. Firstly, complex filtering algorithms must be designed to smooth the original voltage measurement [30]. Secondly, the IC peak position is particularly sensitive to noise [23], which further reduces the effectiveness of the features. Finally, the IC peak position will gradually shift with the battery aging. To obtain the IC features under all aging cycles, it is necessary to sample a large voltage range, which is time-consuming.

For the convenience of comparative analysis, the model complexity of several state-of-the-art SOH estimation methods is listed in Table 1. The n represents the number of cycles of battery degradation data; data volume represents the length of the voltage segment required for battery charging. In summary, the current data-driven methods have some common shortcomings. First, it is difficult to obtain the selected characteristics to characterize battery capacity degradation, and most methods need tedious preprocessing steps for training data. Second, the structure of the model is complex, and the parameters are difficult to adjust in some methods. All these factors further increase the training time of the model, making it challenging to deploy online. Third, most methods require a large charging voltage segment, and the long data sampling time affects the evaluation of battery SOH. Finally, these methods estimate SOH using one or several fixed voltage segments without discussing the universality of the voltage range.

Additionally, previous work has seldom studied the SOH estimation performance in the whole battery SOC range. In the real SOH estimation scenario, the terminal voltage or SOC of the battery is random. When the initial voltage of the battery is not within the required fixed voltage range, an additional charging or discharging operation is required, which is usually a very time-consuming process and may increase the estimation error. Especially in battery screening or health assessment scenarios, this limitation will greatly increase the waiting time of users. In addition to the universality of voltage range, the adaptability of SOH estimation to different working environments and conditions should also be discussed.

To solve these problems, a new constant current charging time (CCCT) based SOH estimation method using the random forest regression (RFR) algorithm is proposed. This method uses less data, features (Input feature size is $(n \times 1)$), and a simpler model structure to achieve higher SOH estimation accuracy. Besides, data preprocessing is not needed in this method, and a fast model training and estimation speed is maintained. Furthermore, considering the respective advantages of the charging voltage curve and IC curve, we try to correspond the direct features from the charging curve to IC peak and obtain encouraging results. The CCCT in the fixed voltage range corresponds to the main peak area of the IC curve was discovered. Although the CCCT feature has been mentioned in Refs. [19,32], the time of the whole CC or CV charging process is used, and charging time is only one of many HIs in these papers. Besides, the charging time feature's importance and significance have not been discussed in Refs. [15,33]. More importantly, due to the lack of mechanism-involved explanation, it is difficult to accept that the duration of some voltage segments is decisive for battery aging. This problem is explained in the subsequent analysis of this paper, while ensuring that the charging time features are fast and easy to obtain (without filtering), the IC peak can also be used to enhance the interpretability of charging time for battery aging. The main contributions of this paper are listed as follows.

1) The corresponding relationship between the CCCT and the main peak area of the IC curve is confirmed, and the effectiveness of the feature is maximally preserved without filtering.
2) The CCCT is used as the only input feature of RFR. Fast and accurate SOH estimation is realized using only a 0.01V or 0.1V charging voltage segment.

Table 1
Model complexity of several SOH estimation methods.

Methods	Input feature size	Feature acquisition difficulty	Data volume	Data preprocessing
M1 [12]	$(n \times 11)$	difficulty	Δ0.3V	Filter
M2 [17]	$(n \times 100)$	ordinary	⊿0.2V	Integral
M3 [19]	$(n \times 8)$	difficulty	⊿0.8V	Filter
M4 [23]	$(n \times 10)$	ordinary	Δ0.2V	Filter
M5 [31]	$(n \times 21)$	difficulty	Δ0.2V	Differential

3) The leave-one-out cross validation method is used to carry out experiments on the Oxford dataset and CALCE dataset. Compared with existing methods, the proposed method owns higher accuracy and efficiency.
4) The universality of SOH estimation is discussed. The experimental results show that the proposed method has great adaptability to different voltage ranges, charging/discharging rates, and temperatures.

The remainder of this paper is organized as follows. Section 2 proposes the overall framework of SOH estimation. The battery capacity degradation is analyzed, and the features are extracted and visualized. The algorithm principle, evaluation criteria, and online SOH estimation are introduced. Section 3 discusses the experimental verification method and results. Finally, the conclusion is given in Section 4.

2. The proposed CCCT based SOH estimation method

The battery CCCT based SOH estimation framework is shown in Fig. 1, consisting of the offline training stage and the online estimation stage. In the offline training stage, the charging process is extracted from the battery degradation dataset, and the characteristics of the battery capacity degradation are analyzed. Then, the charging time feature is extracted to describe the battery aging process. The Random Forest regression model is used to predict the battery SOH. Finally, the verification effect, including time and accuracy information, is stored in the effect evaluator. The trained models are stored in the model database, convenient for calling quickly in online estimation. In the online estimation stage, the estimation decision is made according to the initial battery state and the effect evaluator. The offline model is used to predict the sampled data to complete the online SOH estimation.

2.1. Battery dataset

The Oxford dataset [23] is chosen as a case study in this paper. The dataset contains the aging data of eight commercial Kokam pouch cells of 740-mAh nominal capacity, labeled as cell 1# to cell 8#. All of them are tested in a hot chamber at 40 °C. All eight cells are discharged by the ARTEMIS urban driving cycle [34] and then charged by 2C constant current to complete the cycle. After every 100 aging cycles, a full charge-discharge cycle is conducted at 1C rate to complete a characteristic test. The battery capacity at the end of each test cycle can be calculated by the ampere-hour integration method.

2.2. Capacity degradation analysis

Fig. 2(a) shows the charging voltage curves of different aging stages of cell 1#. The capacity voltage curve and incremental capacity curve can be obtained by ampere-hour integral and Gaussian filter, respectively, as shown in Fig. 2(b). With the aging of the battery, the charging curve gradually shifts to the left, while the incremental capacity peak shifts to the lower right corner. Riviere et al. [35] claimed that this trend reflects the increase of the battery internal resistance. The charging curve is densely distributed at around 3.81 V terminal voltage, which is consistent with the position of IC peak (Fig. 2(b)). This region is where the voltage changes most slowly, which is strongly related to the specific electrochemical reaction inside the battery.

If the same voltage range is taken in each aging cycle, the charging capacity of the battery in this range is equal to the product of charging current and charging time:

$$C_{charge} = I \cdot t_{charge\_time} \tag{1}$$

When the battery adopts constant current charging mode, and the charging rate is fixed, I in (1) is a constant, then the charging

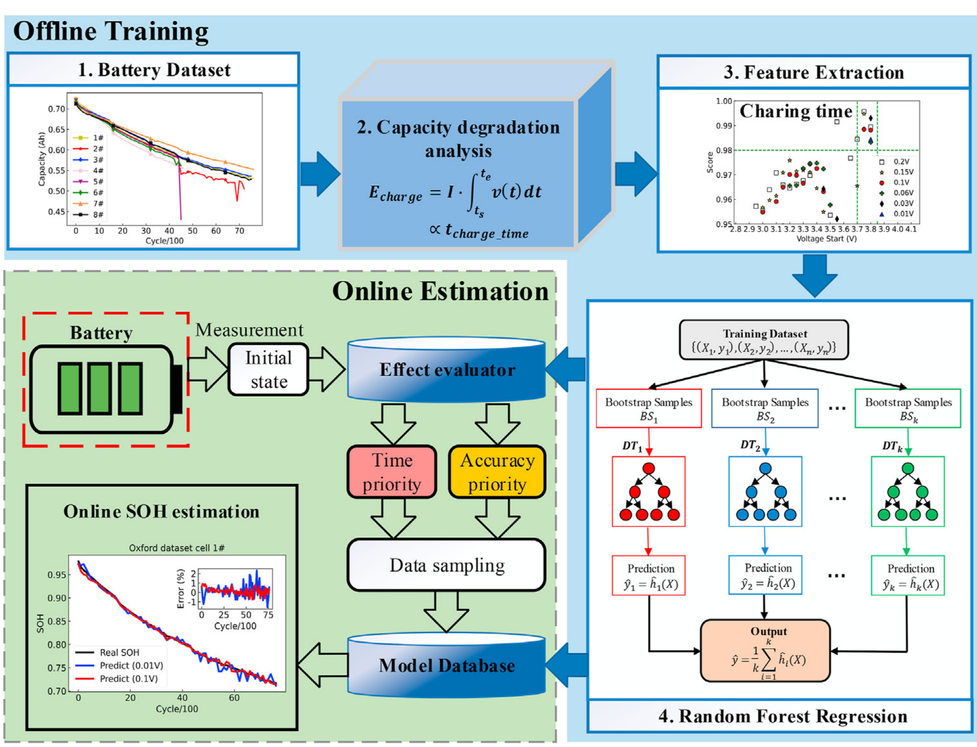

Fig. 1. The overall framework for the CCCT based SOH estimation of lithium-ion battery.

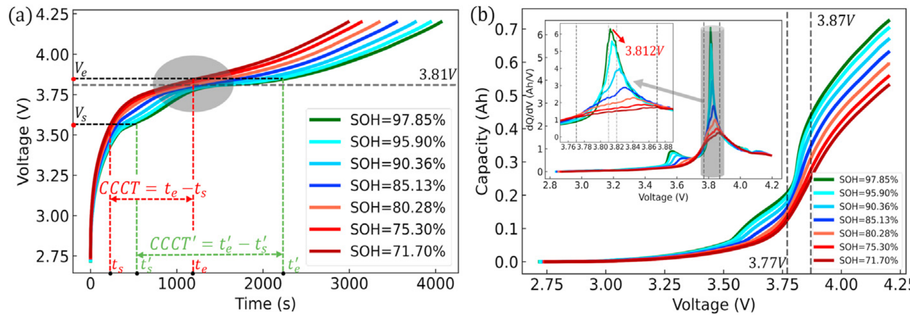

Fig. 2. Battery capacity degradation analysis (Oxford dataset cell 1#). (a) Charging voltage curves at different aging stages. (b) Capacity voltage curve and incremental capacity curve.

capacity in the equal charging voltage range is only related to the charging time (t_{charge_time}). Fortunately, the IC main peak area represents the charging capacity of the fixed voltage range, which has been proved to be an optimal IC feature in Ref. [25]. It means that the charging time can perfectly replace the IC peak area as a more reliable feature. The charging time feature has many advantages, and it is direct and easy to obtain. Usually, it only needs to record the time difference between the starting voltage and the ending voltage, and does not care about the intermediate data. Moreover, it does not need any filtering processing (obtaining IC curve) and integration operation (calculating peak area), to avoid the problem of amplifying voltage measurement noise. Therefore, the charging time in the fixed voltage range is an excellent health indicator, called the constant current charging time (CCCT). The corresponding SOH estimation method is called the CT method in this paper. Fig. 2(a) shows that the SOH decreases with the decrease of the CCCT.

On the other hand, the aging behavior of batteries can be described from the perspective of energy. Within the equal voltage range of battery charging, the chargeable energy E_{charge} is the product of the area under the charging curve and the current. The trapezoidal integral formula is used for approximation, E_{charge} can be expressed as (2) in a relatively small voltage range.

$$E_{charge} = I \cdot \int_{t_s}^{t_e} V(t)dt \approx I \cdot \frac{V_s + V_e}{2} \cdot (t_e - t_s) \propto t_{charge\_time} \tag{2}$$

where I is the current constant in the constant current charging process, and t_s, t_e, V_s, V_e are the starting and ending point of charging time and voltage, respectively. Obviously, E_{charge} can also be used as a health indicator. Within a fixed charging voltage range, the rechargeable energy is directly proportional to the charging time, so the CCCT is used instead of E_{charge}. Thus, the problem of error accumulation caused by the integral formula is solved, and the burden of the acquisition system is significantly reduced.

2.3. Feature extraction

When the specific location of IC peak is not known in advance, the effectiveness of CCCT depends on the selection of voltage range, as shown in Fig. 2(a). A 0.05 V interval is used, and the starting voltage and segment length are set to (2.8, 2.85, …, 4.1) V and (0.01, 0.03, 0.06, 0.1, 0.15, 0.2) V, respectively. Because the cell 1# to be estimated cannot be used in the feature extraction process, the CCCT of all aging cycles of cells 2# to 8# for each voltage segment is calculated, taken as the independent variable. The corresponding SOH value of each cycle is used as the dependent variable. These data points are randomly divided into two parts, and 80% of data are inputted into the Decision-tree regression model for model training and the rest for testing. Decision-tree is the basic model of random forest, and it has a simpler principle and faster training speed, so it is very suitable to select the optimal voltage range here. For details, please refer to Section 2.4. The model determination coefficient R^2 is used as the effect index to evaluate the test data. The closer the R^2 score is to 1, the better the independent variable can explain the dependent variable in regression analysis.

Through experiments, it is found that when the starting voltage is [3.7, 3.85] V, the R^2 of some voltage segments are significantly higher than that of other voltage segments. Then, the voltage interval is reduced to 0.01 V, and the experimental results are shown in Fig. 3(a). When the starting voltage is 3.77 V, the 0.1 V voltage segment (red dot) achieves the highest R^2 score of 0.9941. However, when the starting voltage is 3.81 V, the 0.01 V voltage segment (blue triangle) achieves the highest R^2 score of 0.9890, which shows the potential for fast SOH estimation. In fact, [3.77, 3.87] V and [3.81, 3.82] V are exactly the position where the main peak of IC curve appears, as shown in Fig. 2(b). The potential to obtain interpretable features only from direct measurement data (charging voltage) is confirmed by this encouraging conclusion. In addition, it can be noted that the length of 0.2 V voltage segment (grey square) achieves the highest R^2 score of all combinations at 3.76 V. Although the feature fitting effect is slightly improved, it takes a huge time to get sampling data, so this paper will not discuss it.

Thus, the best voltage range [3.77,3.87] V and [3.81,3.82] V are chosen, respectively, to meet the accuracy and rapidity requirements of SOH estimation. This conclusion is applied to all cells on the Oxford dataset, and the CCCT features are visualized as shown in Fig. 3. It can be noted that the [3.77,3.87] V CCCT features (Fig. 3(b)) are highly linearly correlated with SOH. In contrast, the relationship between [3.81,3.82] V CCCT features (Fig. 3(c)) and SOH is similar to a quadratic convex curve. These two kinds of obvious mapping relations are conducive to the completion of SOH estimation. In addition, both in Fig. 3(b) and (c), when the SOH of the cell 2# (red star) is lower than 0.75, some data deviate from the curve, which is consistent with the abnormal capacity degradation behavior of the cell 2# reflected in the dataset (Fig. 1). It will directly affect the SOH estimation performance of cell 2#. On the other hand, the charge time range of 0.1 V CCCT is about 600 s−1300 s, while the charge time range of 0.01 V CCCT can be reduced to 50 s−300 s. Thus, completing battery SOH estimation within 5 min in the actual test is exploited. The CCCT features of [3.77,3.87] V and [3.81,3.82] V are directly used as the input of the random forest regression model to realize the accurate and fast SOH estimation and directly apply it to all cells on the Oxford dataset to prove the generalization performance of the proposed method.

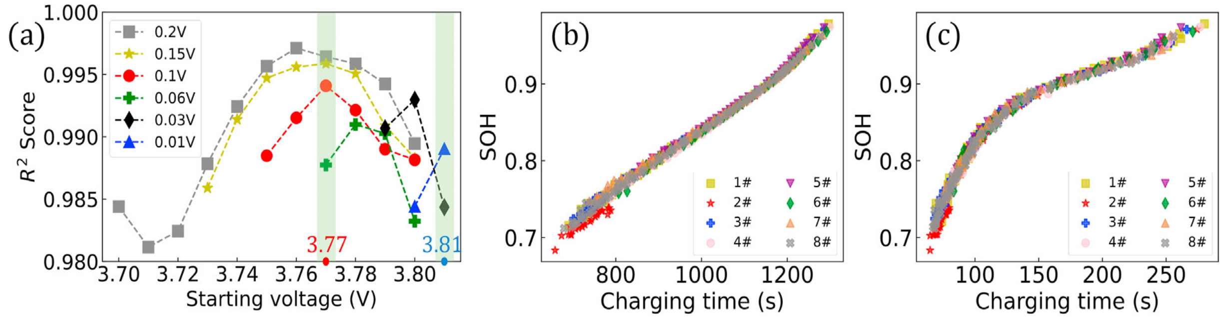

Fig. 3. Feature selection and visualization. (a) The R^2 score of test data in different charging voltage ranges. (b) The relationship between CCCT features ([3.77, 3.87] V) and SOH of all cells. (c) [3.81, 3.82] V CCCT features.

2.4. Random forest regression algorithm

Random forest (RF) has excellent comprehensive performance [10] and is reported to be an effective SOH prediction algorithm [15]. Considering the interpretability, generalization ability, parameter adjustment, and training speed, RF is used to establish a regression prediction model between CCCT and SOH in this paper. RF is an ensemble learning method based on bagging (bootstrap aggregation), first proposed by Breiman [36]. Its core is bootstrap, i.e., the concept of putting back sampling. Several bootstrap samples are obtained from the whole dataset using bootstrap and column sampling technology, and then the basic model is trained by these subsamples. The prediction average value is taken as the output of the final model, thereby effectively reducing the prediction variance and avoiding the overfitting phenomenon of the basic model [17,37].

The basic model chosen is the decision-regression tree, where the number of trees is 100 (RF default parameter). The decision-tree is a nonparametric statistical model first proposed by Breiman et al., in 1984 [38]. It takes mean-square error (MSE) as the basis of node partition. The process of establishing a regression tree is as follows.

1) The set of possible values of the independent variable's characteristic space $(x^{(1)}, x^{(2)}, \ldots, x^{(p)})$ is divided into j nonoverlapping regions $R_1, R_2, \ldots, R_j$.
2) Selecting the best segmentation feature j and the best s on the feature, traversing the feature j and traversing the segmentation point s after fixing j, and selecting the smallest (j, s) that makes (3) hold.

$$\min_{j,s}\left[\min_{c1}\sum_{x_i\in R_1(j,s)}(y_i-c_1)^2+\min_{c2}\sum_{x_i\in R_2(j,s)}(y_i-c_2)^2\right] \tag{3}$$

3) Splitting characteristic space according to (j, s).

$$R_1(j,s)=\left\{x\middle|x^j\le s\right\}, R_2(j,s)=\left\{x\middle|x^j\ge s\right\} \tag{4}$$

$$\widehat{c}_m=\frac{1}{N_m}\sum_{x\in R_m(j,s)}y_i\ ,\ m=1,2 \tag{5}$$

4) Repeating steps 2) and 3) until the stop condition is satisfied, i.e., the number of samples in each region is less than or equal to a certain threshold (parameter: min_samples_split).
5) The feature space is divided into j different regions to generate a regression tree.

$$f(x)=\sum_{m=1}^{j}\widehat{c}_m I(x\in R_m) \tag{6}$$

Random forest combines the decision-tree generated by the bagging algorithm to improve the generalization performance of the model. For a detailed introduction of decision-tree regression, please refer to Ref. [38]. In this paper, the random forest model is trained and tested on a desktop computer with Python 3.7. The specific configuration is: Intel (R) Core (TM) i3-10100 CPU @ 3.60 GHz, 16 GB RAM.

2.5. Online estimation

The SOH estimation of lithium-ion batteries in different initial states is quite different, and there is a contradiction between time and accuracy. In general, the shorter the test time means fewer data collected, which will increase the estimation error. On the contrary, high precision estimation usually requires many test data, which inevitably increases the sampling time. Therefore, it is necessary to balance time and accuracy. As shown in Fig. 1, the SOH estimation effects of voltage segments with different lengths are evaluated by the effect evaluator during offline training. The test time and estimation accuracy are recorded in advance to guide online estimation. This intelligent decision makes the SOH estimation more flexible and efficient. On the online estimation, the battery's initial state is obtained through measurement. A time priority or accuracy priority strategy is provided by the effect evaluator for the starting voltage for users to choose. Then the corresponding constant current charging operation and data collection are performed. The corresponding model from the model database is called to predict the test data and complete the online SOH estimation. Finally, the decision-making behavior is guided according to the estimation results, such as determining the battery health level or judging whether the battery needs to be replaced.

In the present paper, in addition to three common evaluation criteria, the mean-absolute error (MAE), the root-mean-square error (RMSE), and the maximum-absolute error (MaxAE), the model determination coefficient R^2 is also used to evaluate the SOH estimation results.

$$R^2=1-\frac{\sum_{i=1}^{n}(\widehat{y}_i-y_i)^2}{\sum_{i=1}^{n}(y_i-\overline{y}_i)^2} \tag{7}$$

where $\widehat{y}_i, y_i, \overline{y}_i$ represent the predicted value, the actual value, and the average value, respectively. n represent the total number of capacity measurements in the test data. R^2 represents the proportion of the variance of the estimated SOH, which is explained by the input in the SOH estimator. The ideal value of R^2 is 1, which can be used to measure the fitting degree of the predicted value to the actual value.

3. Experimental validation and analysis

To highlight the advantages of the CT method in this paper, under the condition of ensuring the same dataset and verification method, the differences between different methods are compared with the same evaluation criteria. On the Oxford dataset, 8 cells for leave-one-out cross validation, i.e., the training data comes from the remaining 7 cells, and the degradation data of the tested cell 1# does not appear in the training set, to reflect the robustness and generalization performance of the model [31].

3.1. The SOH estimation results

Fig. 3(b) shows a linear relationship between the CCCT feature of [3.77, 3.87] V and SOH during the battery's charging process. Therefore, we use a straight line to fit the data of cells 2# to 8# and apply it to cell 1#. The results are shown in Fig. 4. Using this line to estimate the SOH of cell 1#, except for the estimation error of the first few cycles is too large, the SOH estimation error of about 90% cycles is less than 1%. The MAE, RMSE, and MaxAE are 0.35%, 0.50%, and 1.70%, respectively. In addition, the R^2 is 0.9954, which indicates that the predicted value can well fit the true value. equation (8) of this line can be used as an empirical formula for SOH estimation, which enhances the interpretability of the proposed SOH estimation method.

$$\hat{y}_{SOH} = 0.0004t_{charge\_time} + 0.4337 \quad (8)$$

On the other hand, through the RF regression, the SOH estimation methods based on [3.81,3.82] V and [3.77,3.87] V are abbreviated as CT1 and CT2, respectively. The experimental results are shown in Fig. 5 (only the results of cell 1# and 2# are shown, other cells are similar). Both two methods can well follow the declining trend of SOH. CT2, represented by the red dot, has a better estimation effect than CT1. CT2 strives for higher estimation accuracy at the expense of acquisition or test time. However, it only uses a 0.1 V charging voltage range, which is smaller than most current SOH estimation methods. The CT1 method can shorten the test time to 300s and maintain a high level of SOH estimation. In addition, when the SOH of cell 2# (Fig. 5(b)) is around 0.75, the capacity drops suddenly, which causes the estimation error to increase. This phenomenon is consistent with the description in Section 2.3. The abnormal point in the test phase (such as the 6900th cycle of cell 2#) may indicate the potential safety hazard of the battery. However, none of the other cells used for training showed this behavior. At this time, the data-driven method may have large estimation inaccuracy, so it is necessary to determine the real health status of the battery through more precise measurement methods. The proposed method will recognize these anomalies according to the threshold conditions for deep health analysis.

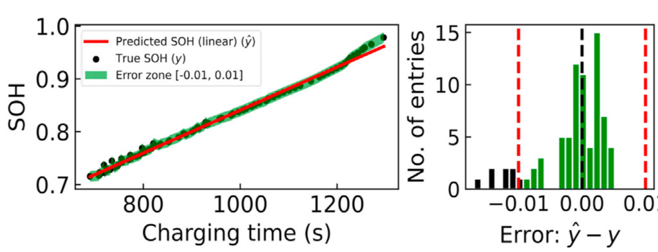

Fig. 4. Linear regression fitting and error diagram of 3.77 V−3.87 V charging process of Oxford cell 1#.

Fig. 6 shows the MAE estimated by SOH of CT1 and CT2. On the 8 cells in Oxford battery dataset, the MAE of two methods is less than 1%, and the average MAE is 0.54% and 0.38%, respectively. When the test time is reduced by 4 times, the average MAE of CT1 is increased by 42.1% compared with CT2. The estimation error of the CT2 method on 8 cells is all smaller than that of the CT1 method, and both methods have the maximum estimation error on cell 2#, which is consistent with the phenomenon observed in Fig. 3(b) and (c).

3.2. Comparison with DT/ICA

The cross validation experiments similar to that in Ref. [31] were performed on the Oxford dataset. The specific values of RMSE and MaxAE of CT1, CT2, and the methods mentioned in Ref. [31] are shown in Table 2. DT and ICA refer to the method of inputting the differential characteristics of battery temperature and incremental capacity peak into SVR for SOH estimation. We have made a full comparison, CT2 based on the voltage range of [3.77,3.87] V has achieved the best results. It has the maximum RMSE of 0.97% and the average RMSE of 0.52% on the 8 cells of Oxford dataset. Compared with the DT and ICA methods, they are reduced by 70.3% and 41.6%, respectively. In addition, its maximum-absolute error is also well controlled, the MaxAE is 3.26% on cell 2#, and there is no loss of control of SOH estimation on cell 5#, unlike DT and ICA methods, in which MaxAE is more than 19%. On the other hand, CT1 based on the voltage range of [3.81,3.82] V has also achieved better estimation results. Its average RMSE is 62.2% and 21.3% lower than DT and ICA methods, respectively. The maximum MaxAE can be controlled within 3.30%. However, the voltage range used in the method in Ref. [31] is [3.6,3.8] V, which means that the same or higher SOH estimation accuracy can be obtained only by using the segment with the length of 0.01 V on the battery charging curves.

Fig. 7(a) shows the comparison of R^2 score of four SOH estimation methods. The average scores of CT1 and CT2 are higher than those of DT and ICA methods, and CT2 has the highest R^2 average score of 0.9937, indicating the superiority of the CCCT feature. The extremely short model training time is another great advantage of CT method. As shown in Fig. 7(b), the proposed method has completed the model training and prediction of 8 cells in 1 s. However, to complete the same task, DT and ICA methods take 508 s and 692 s, CT reduces the time by about 500−700 times. In addition, in the actual SOH estimation scenario of battery, the acquisition time of test data is more important. The CT1 can complete data acquisition and SOH estimation in 300 s, which is not achieved by most SOH estimation methods at present. The 300 s is even shorter than the model training time of DT or ICA method in Ref. [31]. From the indicators in Table 2 and Fig. 7, compared with the DT or ICA methods, the proposed method can overcome the differences between different cells, and there is no huge SOH estimation error on individual cells. Therefore, the CT estimation method not only has high accuracy, fast training time but also has good robustness and strong generalization ability.

3.3. Validation on the CALCE dataset

To further validate the generalization of the proposed SOH estimation method, the proposed method is applied to the two different lithium-ion batteries on CALCE dataset [39] in this section. The overview of three battery datasets is listed in Table 3. It can be noted that they have different manufacturing parameters,

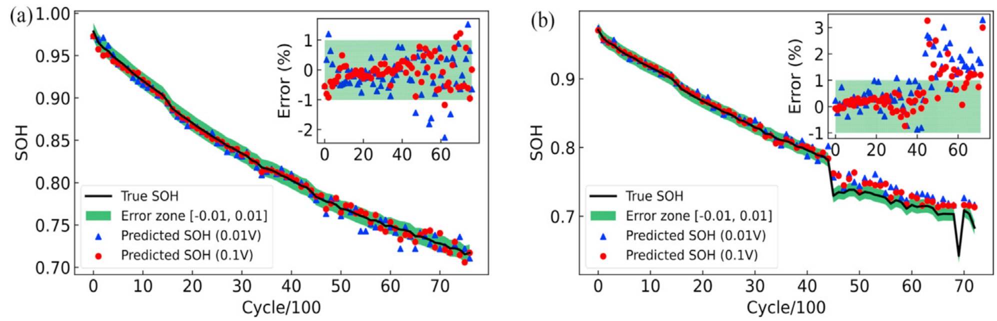

Fig. 5. SOH estimation results based on Random Forest regression. Oxford dataset (a) cell 1# and (b) cell 2#.

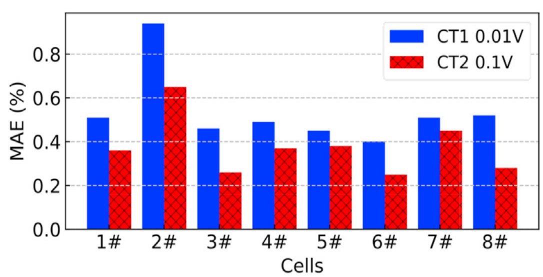

Fig. 6. MAE comparison chart of CT1 and CT2 estimation methods.

Table 3
The overview of three battery datasets.

Dataset	Oxford	CALCE CS2	CALCE CX2
Number of cells	8 (1#-8#)	4 (35#-38#)	4 (35#-38#)
Form factor	Pouch	Prismatic	Prismatic
Cathode	$LiCoO_2$/$LiNiMnCoO_2$	$LiCoO_2$	$LiCoO_2$
Capacity Rating	740 mAh	1100 mAh	1350 mAh
Voltage ranges	2.7 V–4.2 V	2.7 V–4.2 V	2.7 V–4.2 V
Charge	CC (2C)	CC-CV (0.5C)	CC-CV (0.5C)
Discharge	Artemis urban driving cycle	CC (1C)	CC (1C*)
Temperature	40 °C	25 °C	25 °C

Note: *the constant current discharge rate of CX2-35 battery is 0.5C.

Table 2
Comparison of RMSE and MaxAE of different SOH estimation methods.

Cells No.	RMSE (%)				MaxAE (%)			
	CT1	CT2	DT [31]	ICA [31]	CT1	CT2	DT [31]	ICA [31]
1	0.67	0.47	1.43	0.65	2.27	1.22	2.79	1.55
2	1.24	0.97	2.81	0.97	3.30	3.26	7.43	2.49
3	0.60	0.36	1.08	0.45	1.75	1.12	2.50	1.16
4	0.61	0.44	1.50	0.41	1.59	1.06	3.15	1.01
5	0.56	0.44	3.62	3.51	1.25	0.93	19.50	22.81
6	0.51	0.37	1.17	0.23	1.35	1.39	3.90	1.08
7	0.73	0.66	1.80	0.40	3.11	3.20	3.61	0.90
8	0.68	0.42	1.39	0.49	1.91	1.79	3.02	1.01
Mean	**0.70**	**0.52**	1.85	0.89	**3.30**	**3.26**	19.50	22.81

Note: the last line of MaxAE shows the maximum value.

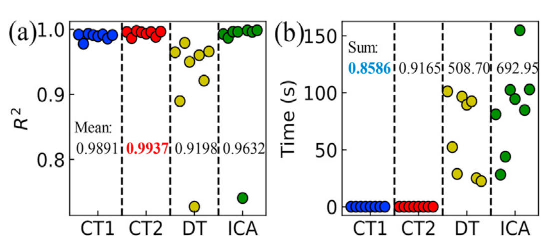

Fig. 7. Comparison of R^2 score and estimation time of different SOH estimation methods. (a) Comparison of the mean R^2 score. (b) Comparison of the total estimation time.

operating temperatures, and charging/discharging cycle forms. The constant current discharge rate of CX2-35 battery is 0.5C, which is different from the other three cells in the same group. These differences provide an interesting perspective for the performance of the proposed CT SOH estimation method.

The feature extraction process in Section 2.3 is repeated, and the optimal voltage ranges of CALCE CS2 and CALCE CX2 are [3.87,3.97] V and [3.89,3.99] V, respectively. The RF regression algorithm is used to estimate the battery SOH, and the cross validation experiments similar to the Oxford dataset are performed. The SOH estimation results of two datasets are listed in Fig. 8 (partial results are shown) and Table 4.

For the four CS2 cells, the MAE and RMSE of SOH estimation are within 1.05% and 1.59%, respectively and the R^2 is higher than 0.9368. It can be noted that except the MaxAE of CS2-38 is 9.83%, the estimation errors of the other three cells are less than 5%, which may be related to the unexpected interruption of 4 days in the CS2-38 aging experiment. For the four CX2 cells, the MAE, RMSE, and MaxAE are within 1.23%, 1.46%, and 4.03%, respectively and the R^2 is higher than 0.9682. As a special validation case, although CX2-35 has a different discharge rate (0.5C) from the other three cells, the SOH estimation error is still within 3.95%.

3.4. Influence of voltage range

In the real SOH estimation scenario, the battery discharge depth rarely reaches 100%, and the battery initial voltage is random, which is difficult for some SOH estimation methods based on fixed charging/discharging voltage interval data. However, as far as we know, most current SOH estimation methods do not consider the universality of voltage range and the estimation performance in all SOCs.

To discuss this problem, an additional experiment was performed for all voltage ranges of the battery in this section. The cross-validation method is adopted, and the CT method based on different voltage ranges is used to estimate the battery SOH. The SOH estimation results for different voltage ranges on the Oxford dataset cell 1# are shown in Fig. 9.

It can be noted that most areas in Fig. 9 are blue. The RMSEs of about 76% of the voltage segment combinations are less than 2%, which shows that the proposed method has a small SOH estimation error in most voltage ranges. Especially when the initial battery voltage is between 3.33 V and 3.81 V, the RMSE can be within 1% by selecting a reasonable voltage segment length. In fact, [3.33, 3.81] V is the region to the left of the incremental capacity peak in Fig. 2(b),

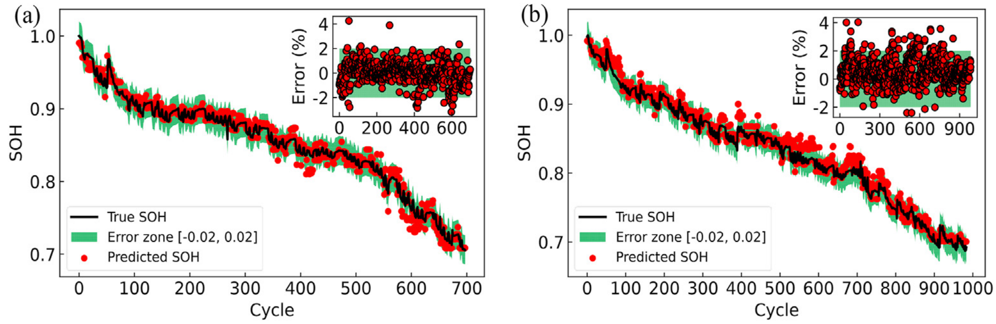

Fig. 8. Partial SOH estimation results based on Random Forest regression. CALCE dataset (a) CS2-37 and (b) CX2-38.

Table 4
The SOH estimation results of CS2 and CX2 on CALCE dataset.

Cells No.	MAE (%)	RMSE (%)	MaxAE (%)	R^2
CS2-35	1.02	1.30	4.94	0.9425
CS2-36	1.00	1.26	4.40	0.9714
CS2-37	0.67	0.89	4.25	0.9805
CS2-38	1.05	1.59	9.83	0.9368
CX2-35	1.23	1.46	3.95	0.9682
CX2-36	0.89	1.09	3.46	0.9782
CX2-37	0.89	1.08	3.69	0.9791
CX2-38	0.81	1.06	4.03	0.9805

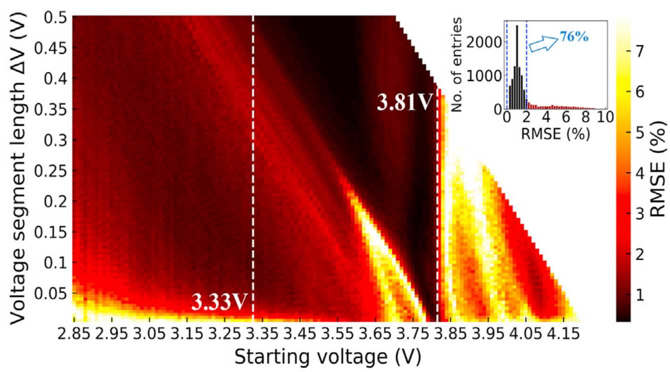

Fig. 9. The SOH estimation error of different voltage ranges on the Oxford dataset cell 1#.

and this part of the IC curves has a high resolution. Therefore, [3.33, 3.81] V is the best starting voltage range for estimating the SOH of this battery. In practical application, it is recommended to estimate SOH when the battery initial voltage is in this range, which can generally obtain more accurate estimation results. On the other hand, in the region where the starting voltage is [3.77, 3.81] V in Fig. 9, the SOH estimation result with RMSE less than 1% can be obtained by using only a voltage segment of 0.05 V or shorter length. In these cases, it is unnecessary to spend more time on constant current charging. When online estimation in these areas, the effect evaluator in Fig. 1 can be used to select the optimal SOH estimation decision. However, when the battery starting voltage is in a high range (e.g. Oxford dataset cell 1# exceeds 3.81V, Fig. 9), the aging curves (Fig. 2(a)) tend to be parallel, and the IC curves are also difficult to distinguish (Fig. 2(b)). At this time, the difference of charging capacity characterized by CCCT is not obvious, which is not enough to indicate the change of SOH, so the proposed method will have a large estimation error in these regions.

As observed from Fig. 9 and Table 3, the proposed method is robust to the battery starting voltage and can provide accurate SOH estimation in most battery SOC states. At the same time, the applied voltage range of the proposed method is universal. It has good adaptability to different constant current charging rates, discharge cycles, operating temperatures, and chemical conditions. In addition, compared with the currently popular methods, the proposed method has lower computational complexity while maintaining generalization, so it is well-suited for online applications.

4. Conclusion

To solve the time-consuming feature extraction, tedious data preprocessing, and high model complexity problems in current mainstream SOH estimation methods, a simpler and faster SOH estimation method has been proposed in this paper. The proposed method takes the charging time in a specific voltage range as the feature and combines it with RF regression algorithm to estimate battery SOH. The relationship between the charging time feature and the peak area of incremental capacity is confirmed, which gives charging time a mechanism-involved explanation. The effectiveness of the feature is not lost because no filtering or integration processing is required. Compared with other methods, the proposed method owns faster training speed and lower computation complexity advantages.

Based on the 0.1 V data of battery constant current charging process, the average MAE and RMSE of 8 cells on Oxford dataset were 0.38% and 0.52%, which were 41.6% lower than the current advanced ICA method, and the MaxAE can be controlled within 3.26%. For the 0.01 V voltage range, the average RMSE of 8 cells on Oxford dataset was reduced by 21.3% compared with the ICA method, and the MaxAE was only 3.30%. Besides, the proposed method greatly reduced the data amount and shortened the sampling test time by about 4 times. Even for new batteries, the SOH estimation can be completed in 300 s. On the other hand, the proposed method has the advantages of small computation, no preprocessing such as filtering and normalization, and fast model training speed. The 8 cells on Oxford Dataset can complete the model training and prediction in 1 s, with great advantages compared with the current mainstream SOH estimation methods. Finally, the generalization of the proposed method was verified on the CALCE dataset. The average RMSE of two different types of batteries were 1.26% and 1.17%, respectively. The additional experimental results demonstrated that the proposed method was adaptable to different voltage ranges, constant current charging rates, discharging cycles, and operating temperatures. Other dimensional features such as temperature and pressure will be integrated into our future work to further improve the accuracy and robustness of SOH estimation.

Credit author statement

Chuanping Lin: Software, Writing – original draft preparation, Formal analysis, Data curation, Validation. **Jun Xu:** Conceptualization, Methodology, Investigation, Writing- Reviewing and Editing, Supervision, Funding acquisition. **Mingjie Shi:** Resources, Data curation, Visualization, Validation. **Xuesong Mei:** Supervision, Reviewing and Editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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